BOTH-WELL STEEL FITTINGS CO., LTD.

NO.303, RENXIN ROAD, RENWU DISTRICT, KAOHSIUNG CITY, TAIWAN, R.O.C. (814020)
TEL : (886)7-3711536, 3710497, 3720260 FAX : (886)7-371-3864, 3713882
WEB SITE : http://www.bothwell.com.tw E-Mail : BW_SalesT1@bothwell.com.tw

An ISO 9001 : 2015 Registered Manufacturer

MILL TEST & INSPECTION CERTIFICATE

ACCORDING TO EN 10204 3.1

CUSTOMER : SILBO INDUSTRIES, INC.
CERT NO. : 201139
ORDER NO. : 70435

INVOICE NO. : 5200015824
L/C NO. :

PAGE : 1
DATE : 05/22/2025
ORIGIN : TAIWAN

ITEM	RAW HEAT NO. HEAT NO.	QTY	DESC/GRADE OR SPECIFICATION NO. CHEMICAL COMPOSITION(%) MECHANICAL PROPERTIES.
357	9SA0180 0I03	100 PCS	HALF CPLG F51/F60 3/4" 3000# NPT C:0.022 Si:0.47 Mn:1.20 P:0.020 S:0.001 Cu:- Cr:22.70 Ni:5.60 Mo:3.30 V:- Nb:- N:0.18 TS(KSI):105.6 YS 0.2%(KSI):78.6 EL(%)G.L.4D:43.6 R.A.(%):82.4 HARDNESS 1(HBW):181 HARDNESS 2(HBW):182 COUNTRY OF MELT AND POUR :TAIWAN

REMARK:
THE MATERIAL SUPPLIED IDENTIFIED AS ASTM A182 -24C, ASME SA182-2023ED, F51/F60
THE FITTING SUPPLIED IDENTIFIED AS ASME B16.11 - 2021
THE PRODUCT SUPPLIED WAS INSPECTED IN ACCORDANCE WITH SPECIFICATION(SURFACE & DIM).: GOOD
STANDARD : CHEMICAL COMPOSITION(%),MECHANICAL PROPERTIES & CE= C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15

	C	Si	Mn	P	S	Cu	Cr	Ni	Mo	V	Cb(Nb)
MIN	-	-	-	-	-	-	22.0	4.5	3.0	-	-
MAX	0.030	1.00	2.00	0.030	0.020	-	23.0	6.5	3.5	-	-
	N	Al	Ti	Zr	CE	TS(KSI)	YS(KSI)	YS 0.2%(KSI)	EL(%)G.L.4D	R of A(%)	HARDNESS(HBW)
MIN	0.14					95		65	25	45	-
MAX	0.20					-		-	-	-	271

HEAT TREATMENT: SOLUTION: 1065℃(1949°F) W.Q.
CONFORMS TO NACE MR0175-2021/ISO 15156-2020 & MR0103/ISO 17945 :2015
TEST METHOD A PERFORMED AND PASSED ASTM G48 CRITERIA
TEST METHOD C PERFORMED AND PASSED ASTM A923 CRITERIA
PED 2014/68/EU ANNEX 1 SECTION 4.3 MATERIAL 0.5HR AIR COOLING
MATERIAL MERCURY FREE AND DETECTED FREE FROM RADIOACTIVE CONTAMINATION.
FERRITE CONTENT ACCORDING TO E562, GRID 16,30 FIELDS,
MAGNIFICATION 500X: 48%

ACCEPTED
BY JR DATE 1/19/26

WE CERTIFY THE ABOVE MENTIONED FITTINGS HAVE BEEN MANUFACTURED, SAMPLED, TESTED, AND INSPECTED IN ACCORDANCE WITH THE SPECIFICATIONS SHOWN

Y.Y. Chang
Q.C. MANAGER

C.L. Ko
INSPECTOR

BW-D0839 REV:4

Job: 089-25 Po: 40301 Line: 006B HT: 0I03

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CORROSION TEST ACCORDING TO ASTM A923 METHOD C ,
24HRS, AT 25 DEGREE C:NO PITTING HAVE BEEN OBSERVED AT 20X
MAGNIFICATION. WEIGHT LOSS 1.4 MG/DM 2 /DAY.

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